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(54) COMMUNICATION DEVICE AND COMMUNICATION CONTROL METHOD

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(57) **ABSTRACT**

A communication device includes: a communication unit configured to perform wireless communication; an access point detection unit configured to detect access points capable of communication connection with the communication unit; and a communication control unit configured to display an access point information image on a display device when communication between the communication unit and a communication terminal is performed in a first frequency band, the access point information image presenting information on the access points detected by the access point detection unit, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in a second frequency band.

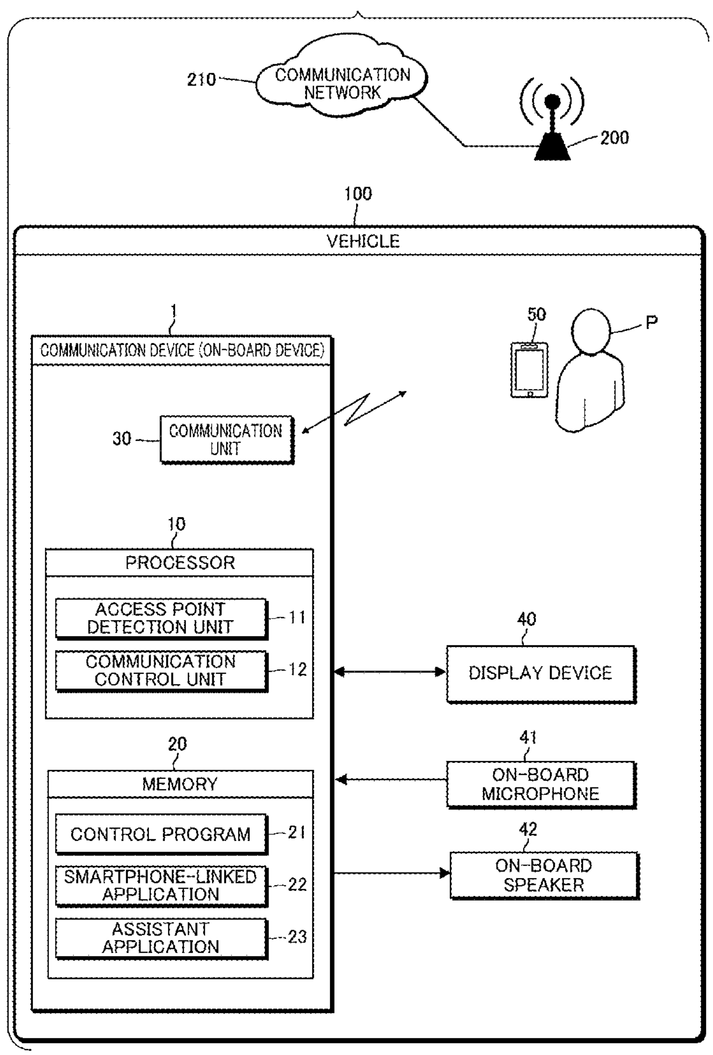


FIG.1

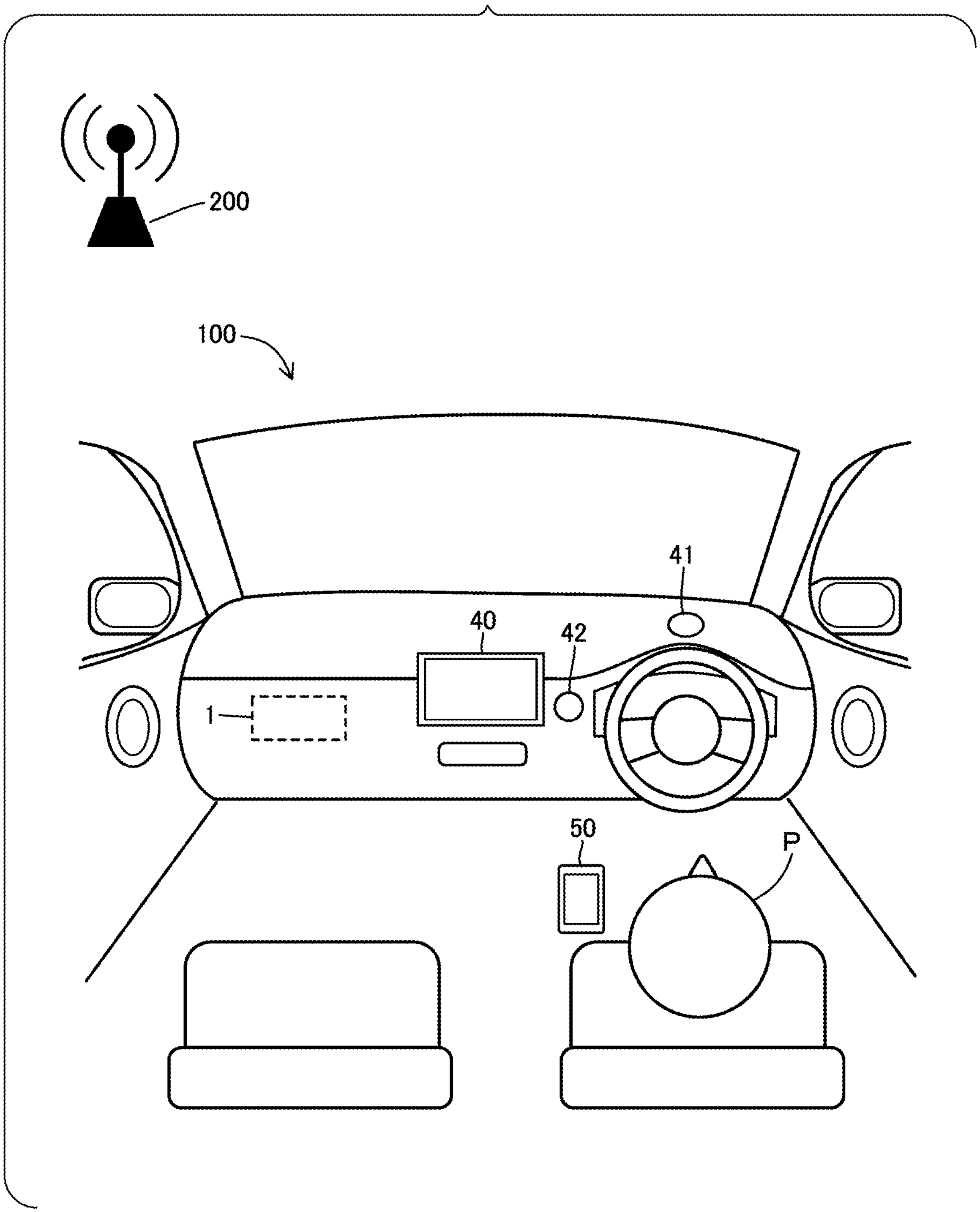


FIG. 2

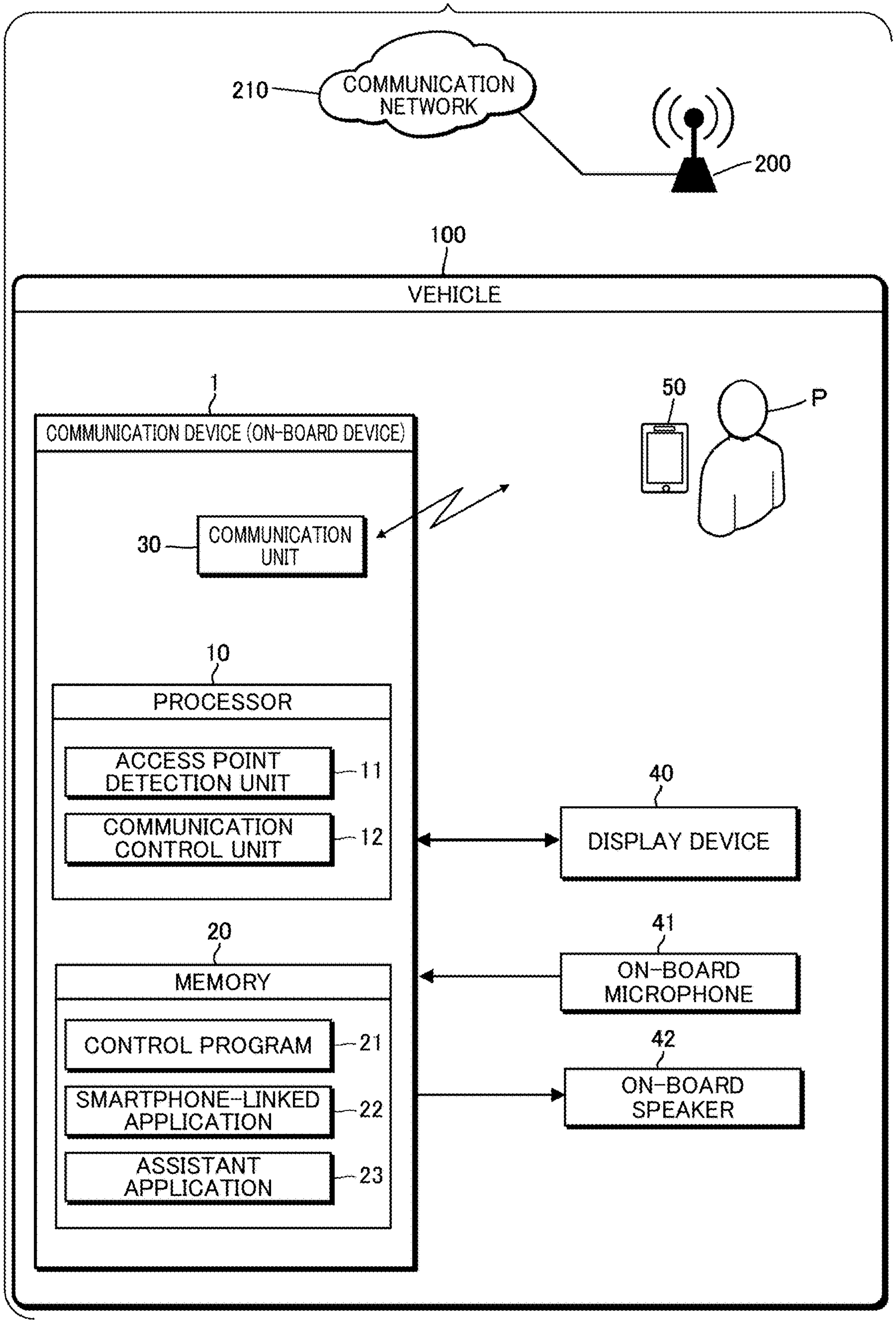

210
COMMUNICATION NETWORK
200
100
VEHICLE
1
COMMUNICATION DEVICE (ON-BOARD DEVICE)
30
COMMUNICATION UNIT
50
P
10
PROCESSOR
ACCESS POINT DETECTION UNIT
11
COMMUNICATION CONTROL UNIT
12
40
DISPLAY DEVICE
20
MEMORY
41
ON-BOARD MICROPHONE
CONTROL PROGRAM
21
42
ON-BOARD SPEAKER
SMARTPHONE-LINKED APPLICATION
22
ASSISTANT APPLICATION
23

FIG.3

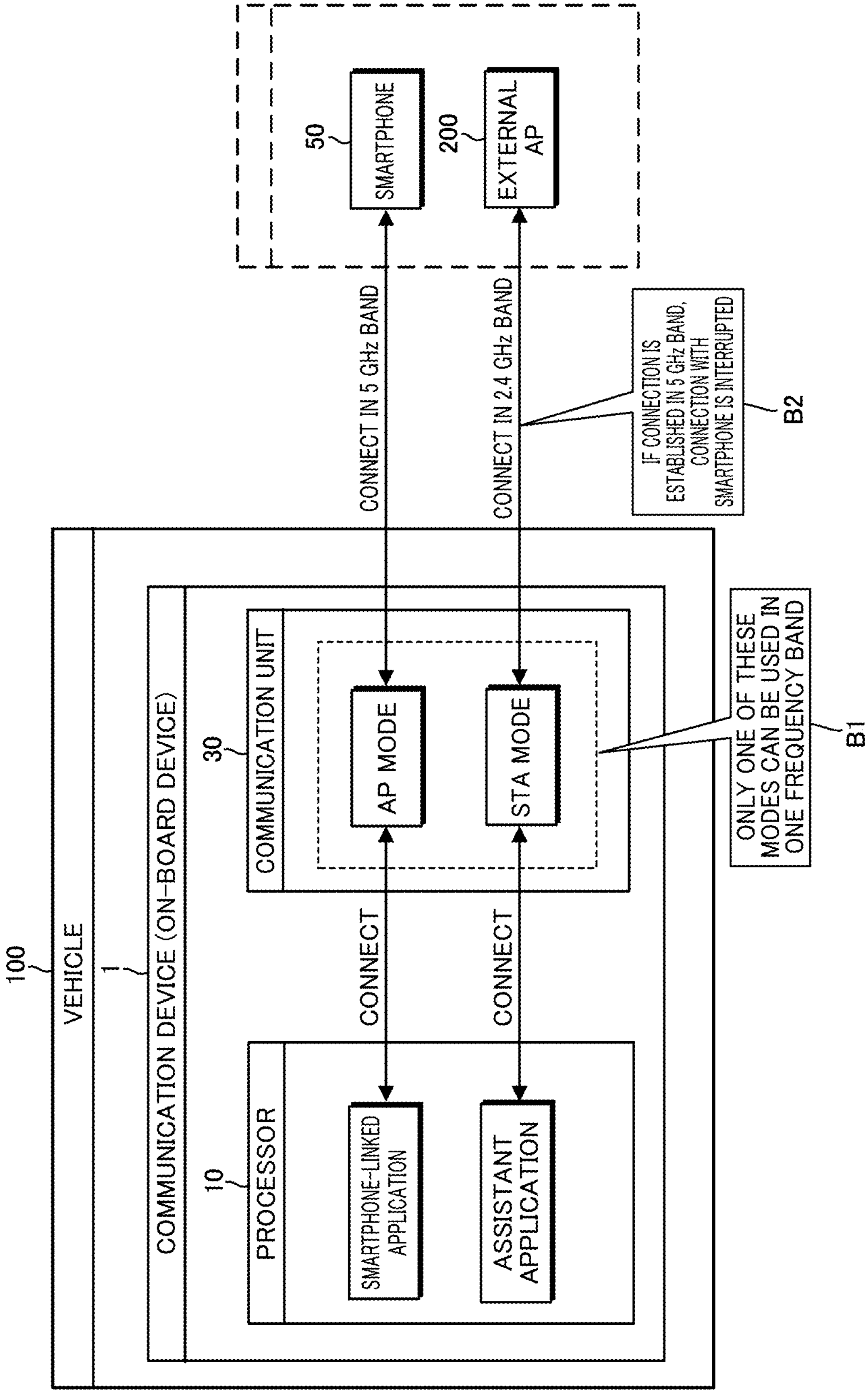

100
VEHICLE
1
COMMUNICATION DEVICE (ON-BOARD DEVICE)
10
PROCESSOR
SMARTPHONE-LINKED APPLICATION
ASSISTANT APPLICATION
CONNECT
CONNECT
30
COMMUNICATION UNIT
AP MODE
STA MODE
ONLY ONE OF THESE MODES CAN BE USED IN ONE FREQUENCY BAND
B1
CONNECT IN 5 GHz BAND
CONNECT IN 2.4 GHz BAND
IF CONNECTION IS ESTABLISHED IN 5 GHz BAND, CONNECTION WITH SMARTPHONE IS INTERRUPTED
B2
50
SMARTPHONE
200
EXTERNAL AP

FIG.4

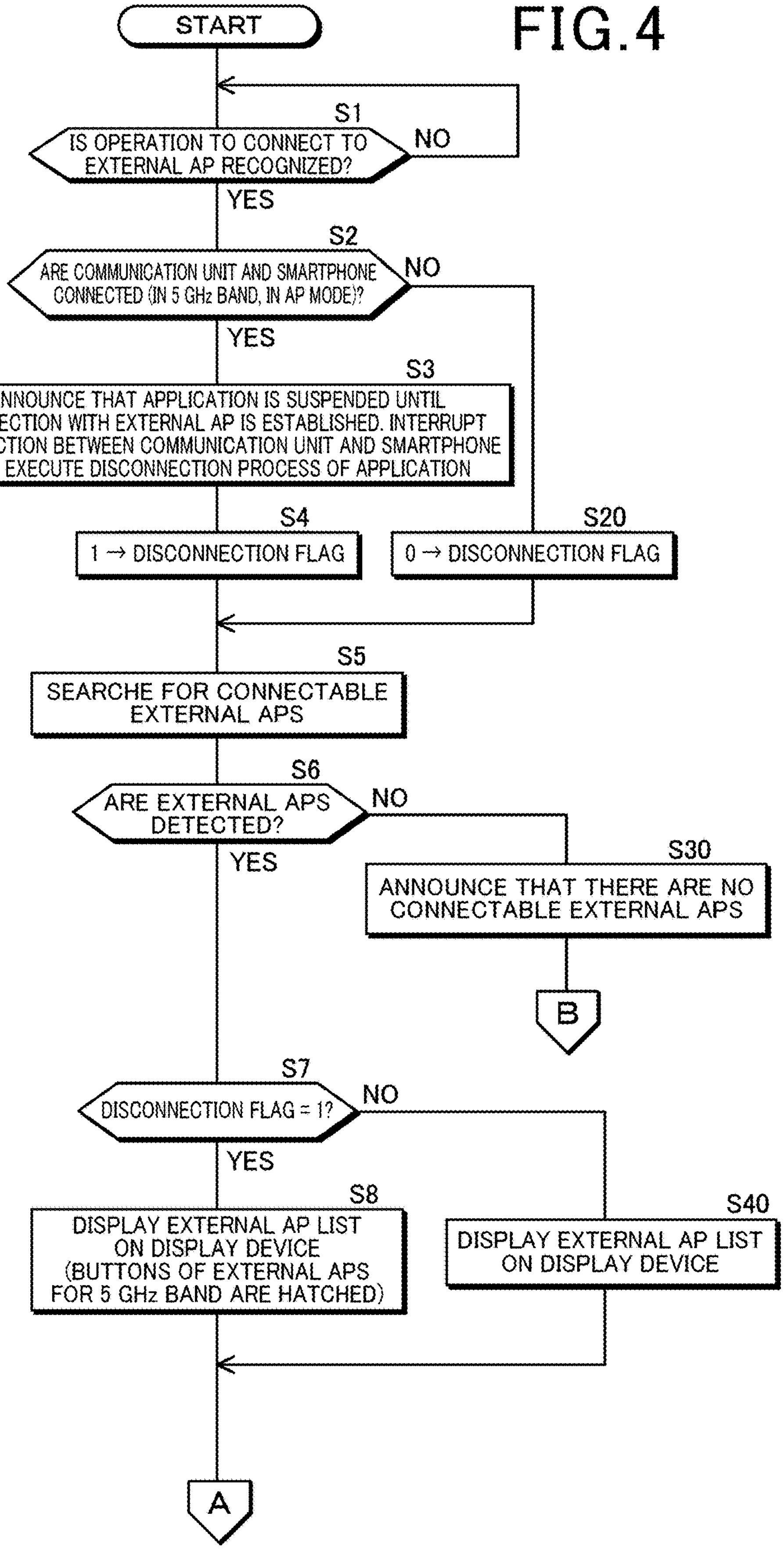

START
S1
IS OPERATION TO CONNECT TO EXTERNAL AP RECOGNIZED?
NO
YES
S2
ARE COMMUNICATION UNIT AND SMARTPHONE CONNECTED (IN 5 GHz BAND, IN AP MODE)?
NO
YES
S3
ANNOUNCE THAT APPLICATION IS SUSPENDED UNTIL CONNECTION WITH EXTERNAL AP IS ESTABLISHED. INTERRUPT CONNECTION BETWEEN COMMUNICATION UNIT AND SMARTPHONE AND EXECUTE DISCONNECTION PROCESS OF APPLICATION
S4
1 → DISCONNECTION FLAG
S20
0 → DISCONNECTION FLAG
S5
SEARCHE FOR CONNECTABLE EXTERNAL APS
S6
ARE EXTERNAL APS DETECTED?
NO
YES
S30
ANNOUNCE THAT THERE ARE NO CONNECTABLE EXTERNAL APS
B
S7
DISCONNECTION FLAG = 1?
NO
YES
S8
DISPLAY EXTERNAL AP LIST ON DISPLAY DEVICE (BUTTONS OF EXTERNAL APS FOR 5 GHz BAND ARE HATCHED)
S40
DISPLAY EXTERNAL AP LIST ON DISPLAY DEVICE
A

FIG.5

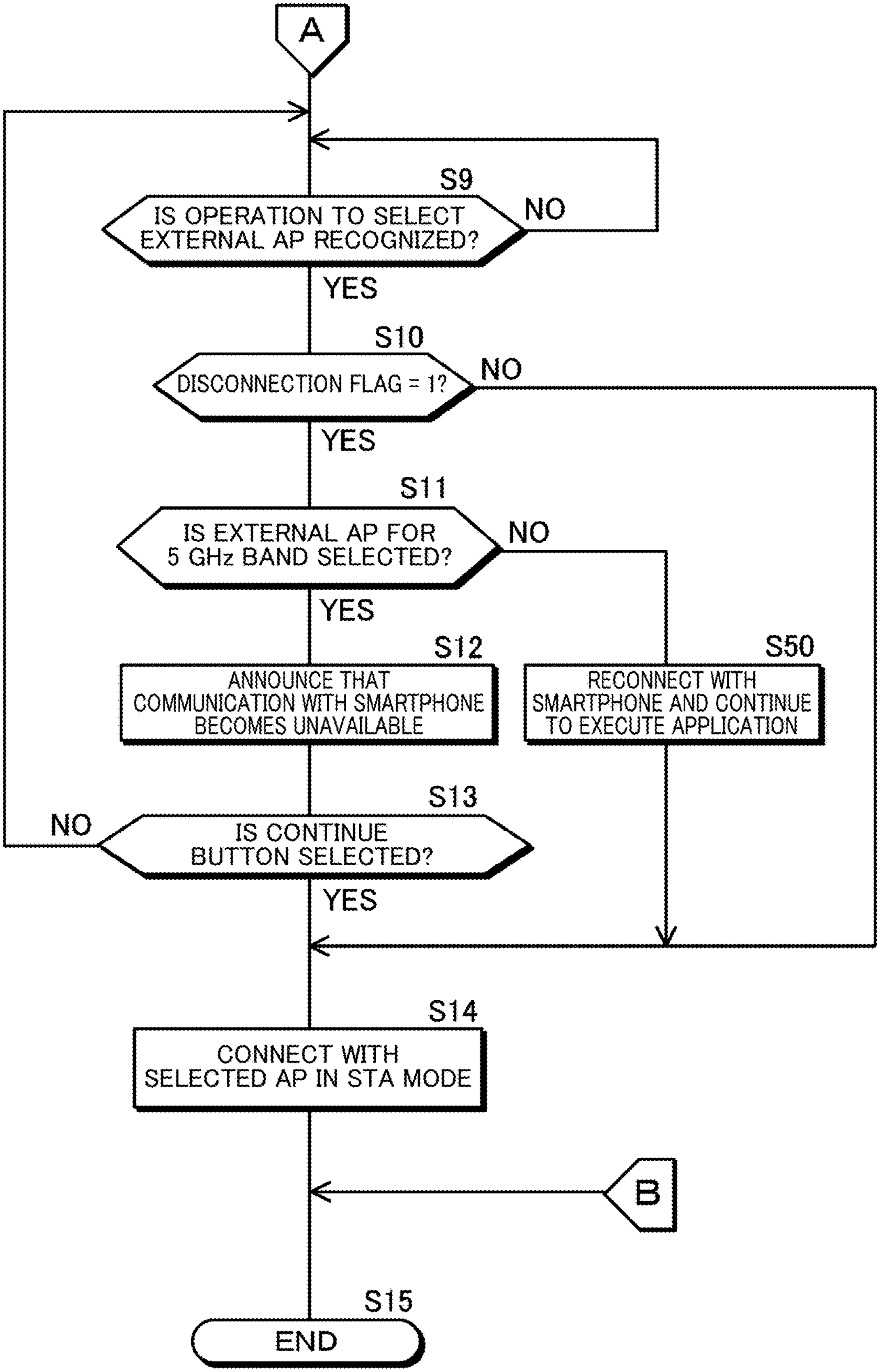

A
S9
IS OPERATION TO SELECT EXTERNAL AP RECOGNIZED?
NO
YES
S10
DISCONNECTION FLAG = 1?
NO
YES
S11
IS EXTERNAL AP FOR 5 GHz BAND SELECTED?
NO
YES
S12
ANNOUNCE THAT COMMUNICATION WITH SMARTPHONE BECOMES UNAVAILABLE
S50
RECONNECT WITH SMARTPHONE AND CONTINUE TO EXECUTE APPLICATION
S13
IS CONTINUE BUTTON SELECTED?
NO
YES
S14
CONNECT WITH SELECTED AP IN STA MODE
B
S15
END

FIG.6

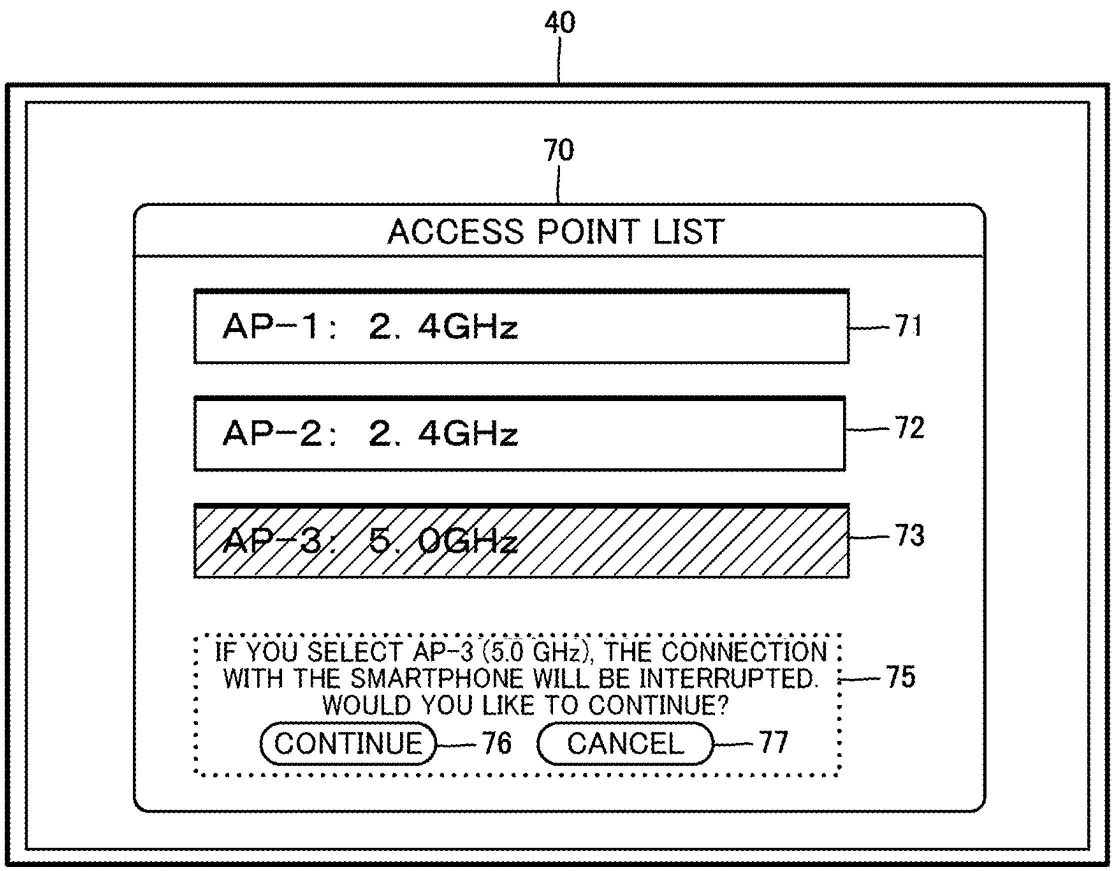
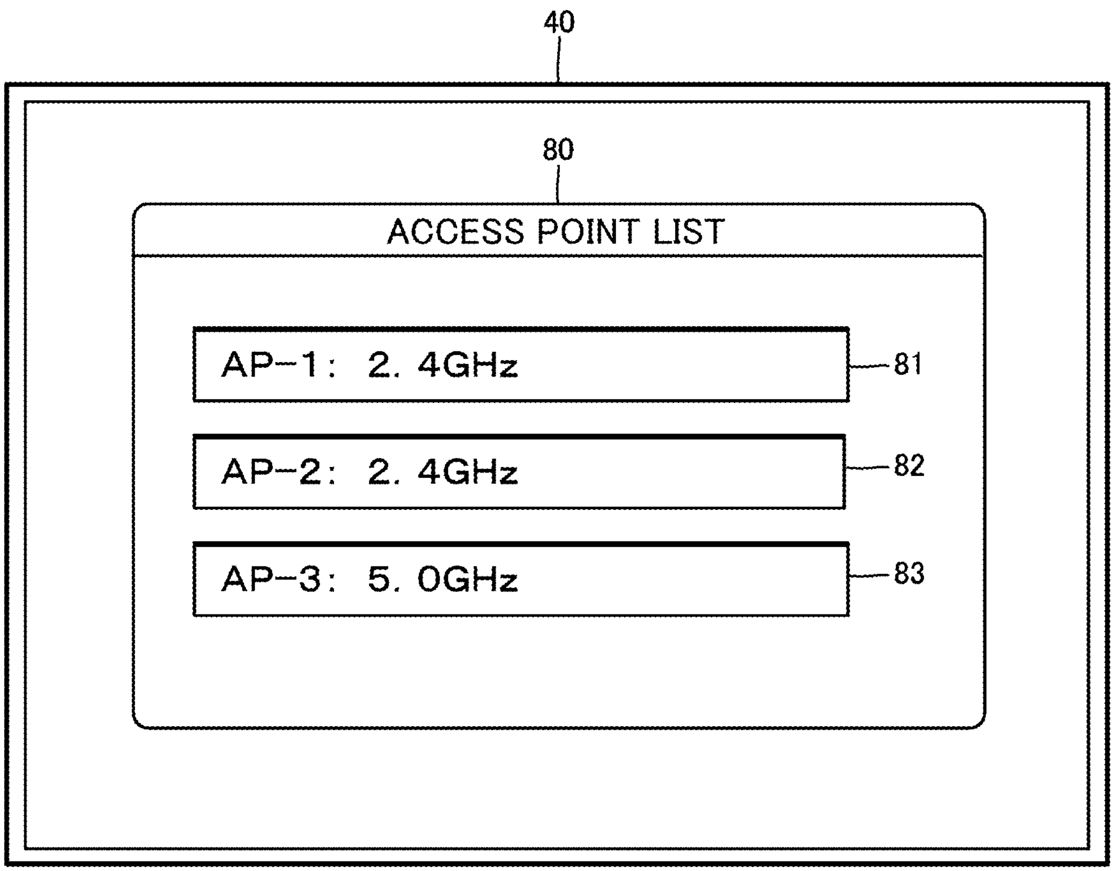

40
70
ACCESS POINT LIST
AP-1: 2. 4GHz
71
AP-2: 2. 4GHz
72
AP-3: 5. 0GHz
73
IF YOU SELECT AP-3 (5.0 GHz), THE CONNECTION WITH THE SMARTPHONE WILL BE INTERRUPTED. WOULD YOU LIKE TO CONTINUE?
75
CONTINUE
76
CANCEL
77

FIG.7



COMMUNICATION DEVICE AND COMMUNICATION CONTROL METHOD

INCORPORATION BY REFERENCE

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-027410 filed on Feb. 27, 2024. The content of the application is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a communication device and a communication control method.

Description of the Related Art

[0003] In the past, a wireless communication system has been proposed, which determines the radio wave condition in the area where an on-board device and a communication terminal outside the vehicle are located, and notifies the on-board device and the communication terminal of a channel and a frequency that allow direct communication between them (see Japanese Patent Laid-Open No. 2009-17101, for example).

[0004] If a communication device used in a mobile object such as a vehicle has a capability to simultaneously communicate with multiple communication targets in different frequency bands, the communication device can, for example, communicate with a communication terminal in a first frequency band and with an access point outside the mobile object in a second frequency band. However, when the communication device and the access point attempt to communicate in the first frequency band while the communication device and the communication terminal are communicating in the same frequency band, inconveniently, the communication between the communication device and the communication terminal is interrupted or the frequency is changed by instructions for connection with the access point without permission because the frequency bands overlap.

[0005] The present invention has been made in view of such background, and the object of the invention is to provide a communication device and a communication control method that can prevent interruption of communication with a communication terminal due to connection with an access point and can prevent change of the frequency by instructions for connection with the access point without permission during the communication with the communication terminal in a mobile object.

SUMMARY OF THE INVENTION

[0006] As a first aspect to achieve the aforementioned object, a communication device is provided, which is used in a mobile object and includes: a communication unit configured to perform wireless communication with communication targets in a first frequency band or a second frequency band; an access point detection unit configured to detect access points capable of communication connection with the communication unit; and a communication control unit configured to control communication between the communication unit and a communication terminal and communication between the communication unit and the access points, and to display an access point information image on a display device used in the mobile object when the communication between the communication unit and the communication terminal is performed in the first frequency band, the access point information image presenting information on the access points detected by the access point detection unit, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in the second frequency band.

[0007] In the above communication device, the communication control unit may be further configured to display the access point information image on the display device when the communication between the communication unit and the communication terminal is performed in the first frequency band and the communication control unit recognizes an operation to connect the communication unit and the access points by an occupant of the mobile object.

[0008] In the above communication device, the communication control unit may be further configured to interrupt the communication between the communication unit and the communication terminal in the first frequency band when the communication between the communication unit and the communication terminal is performed in the first frequency band and the communication control unit recognizes an operation to connect the communication unit and the access points by the occupant, and the access point detection unit may be further configured to detect the access points capable of communication connection with the communication unit by performing communication in the first frequency band and the second frequency band while the communication between the communication unit and the communication terminal is interrupted.

[0009] In the above communication device, the communication control unit may be further configured to start communication between the communication unit and the second access point in the second frequency band and resume the communication between the communication unit and the communication terminal in the first frequency band when the communication control unit recognizes an operation to connect to the second access point by an occupant of the mobile object in response to display of the access point information image on the display device.

[0010] In the above communication device, the communication control unit may be further configured to output a communication unavailability announcement that communication with the terminal device becomes unavailable from an announcement device used in the mobile object when the communication control unit recognizes an operation to connect to the first access point by an occupant of the mobile object in response to display of the access point information image on the display device.

[0011] In the above communication device, the communication control unit may be further configured to output a recommended frequency band announcement recommending connection with the second access point from an announcement device used in the mobile object when the communication control unit recognizes an operation to connect to the first access point by an occupant of the mobile object in response to display of the access point information image on the display device.

[0012] In the above communication device, the communication control unit may be further configured to display on the display device as the access point information image an image in which only information on the second access point is presented and information on the first access point is hidden.

[0013] In the above communication device, the communication control unit may be further configured to display on the display device as the access point information image an image in which information on the second access point is presented preferentially over information on the first access point.

[0014] In the above communication device, the communication control unit may be further configured to change a display style of the access point information image according to an operation to set the display style of the access point information image by an occupant of the mobile object.

[0015] As a second aspect to achieve the aforementioned object, a communication control method is provided, which is performed by a communication device used in a mobile object and including a communication unit configured to perform wireless communication with communication targets in a first frequency band or a second frequency band, the method including: an access point detection step of detecting access points capable of communication connection with the communication unit; and a communication control step of controlling communication between the communication unit and a communication terminal and communication between the communication unit and the access points, and displaying an access point information image on a display device used in the mobile object when the communication between the communication unit and the communication terminal is performed in the first frequency band, the access point information image presenting information on the access points detected in the access point detection step, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in the second frequency band.

[0016] The communication device and the communication control method described above can prevent interruption of communication with a communication terminal due to connection with an access and can prevent change of the frequency by instructions for connection with the access point without permission during the communication with the communication terminal in a mobile object.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. **1** is an illustration of a configuration of a vehicle equipped with a communication device;

[0018] FIG. **2** is a block diagram of the communication device;

[0019] FIG. **3** is an illustration of a mode of connection of a communication target that is a smartphone and an external access point;

[0020] FIG. **4** is the first flowchart of a process of connection with an external access point;

[0021] FIG. **5** is the second flowchart of the process of connection with an external access point;

[0022] FIG. **6** is an illustration of a first example display of an access point information image; and

[0023] FIG. **7** is an illustration of a second example display of the access point information image.

DETAILED DESCRIPTION OF THE INVENTION

1. Configuration of Vehicle Equipped With Communication Device

[0024] The configuration of a vehicle **100** equipped with a communication device **1** in the present embodiment will be described with reference to FIG. **1**. The communication device **1** is configured as part of the functionality of an on-board device installed in the vehicle **100** and is connected to a display device **40**, an on-board microphone **41**, and an on-board speaker **42**. The vehicle **100** corresponds to a mobile object in this disclosure. The display device **40** is a touch panel with a touch sensor arranged on the surface of a flat display such as an LCD panel and an OLED panel. The display device **40** and the on-board speaker **42** correspond to an announcement device in this disclosure.

[0025] The communication device **1** performs wireless communication in the 5 GHz band with a smartphone **50** used by an occupant P of the vehicle **100** and wireless communication in the 5 GHz or 2.4 GHz band with an external access point **200** located around the vehicle **100** and connected to a communication network **210**. The communication device **1** performs wireless communication in compliance with, for example, the Wi-Fi® standard. The 5 GHz band corresponds to a first frequency band in this disclosure, and the 2.4 GHz band corresponds to a second frequency band in this disclosure. The first frequency band in this disclosure may be a frequency band other than the 5 GHz band, and the second frequency band in this disclosure may be a frequency band other than the 2.4 GHz band. In the following, the external access point is also referred to as an external AP.

[0026] The smartphone **50** corresponds to a communication terminal having a processor in this disclosure. The communication terminal in this disclosure includes a cell phone, tablet terminal, etc., in addition to a smartphone. The communication device **1** detects connectable external APs **200**, displays information on the external APs **200** on the display device **40**, and determines to which external AP **200** the communication device **1** connects according to touch operation on the display device **40** by the occupant P.

2. Configuration of Communication Device

[0027] The configuration of the communication device **1** will be described with reference to FIG. **2**. The communication device **1** is a control unit including a processor **10**, a memory **20**, and a communication unit **30**. The communication unit **30** includes a transmitter and a receiver and performs wireless communication in the 5 GHz band with the smartphone **50** and wireless communication in the 5 GHz or 2.4 GHz band with the external AP **200**.

[0028] The memory **20** stores a control program **21** for the communication device **1**, a smartphone-linked application program **22**, and an assistant application program **23**. The smartphone-linked application enables applications running on the smartphone (e.g., a car navigation application) to be used on the display device **40**. The assistant application enables the use of a virtual assistant that supports speech-based operation of the vehicle **100** by the occupant P. The processor **10** functions as an access point detection unit **11** and a communication control unit **12** by reading and executing the program **21**.

[0029] The process performed by the access point detection unit **11** corresponds to the access point detection step in the communication control method of the present disclosure, and the process performed by the communication control unit **12** corresponds to the communication control step in the communication control method of the present disclosure.

[0030] The access point detection unit **11** detects external APs **200** that exist around the vehicle **100** and can be connected to the communication unit **30**. The communication control unit **12** controls wireless communication between the communication unit **30** and communication targets (in the examples in FIGS. **1** and **2**, the smartphone **50** and the external AP **200**).

[0031] The processor **20** executes the smartphone-linked application program **22** to enable the use of the smartphone-linked application and executes the assistant application program **23** to enable the use of the assistant application, as shown in FIG. **3**. The smartphone-linked application connects the communication unit **30** and the smartphone **50** in AP mode (access point mode) in the 5 GHZ band. The assistant application connects the communication unit **30** and the external AP **200** in STA mode (station mode) in the 2.4 GHz or 5 GHz band. FIG. **3** illustrates a situation where the communication unit **30** is connected to the external AP **200** in the 2.4 GHz band.

[0032] As shown in the bubble B**1**, the communication unit **30** can use only one of AP mode and STA mode in the same frequency band. Thus, as shown in FIG. **3**, when the occupant P performs an operation to connect to the external AP **200** in STA mode in the 5 GHz band while the communication unit **30** and the smartphone **50** are connected in AP mode in the 5 GHz band, inconveniently, the communication between the communication unit **30** and the smartphone **50** is cut off and the smartphone-linked application that has been used is no longer usable.

[0033] The communication device **1**, therefore, performs the following connection process with the external AP **200** to prevent the communication between the communication unit **30** and the smartphone **50** from being interrupted due to overlap between the frequency bands used for connection with the smartphone **50** and for connection with the external AP **200**.

3. Connection Process With External AP

[0034] The flowchart shown in FIGS. **4** and **5** describes the procedure of the connection process with the external AP **200** executed by the access point detection unit **11** and the communication control unit **12**. In step S**1** in FIG. **4**, when the communication control unit **12** recognizes an operation to connect to the external AP **200** (e.g., an operation of the communication setting menu displayed on the display device **40**) by the occupant P, the process proceeds to step S**2**.

[0035] In step S**2**, the communication control unit **12** determines whether or not the communication unit **30** and the smartphone **50** are connected (in AP mode in the 5 GHz band). The communication control unit **12** advances the process to step S**3** if the communication unit **30** and the smartphone **50** are connected, or to step S**20** if the communication unit **30** and the smartphone **50** are not connected.

[0036] In step S**3**, the communication control unit **12** announces to the occupant P that the application (smartphone-linked application) is suspended until the connection with the external AP **200** is established, either by display on the display device **40** or by audio output from the on-board speaker **42**. The communication control unit **12** also interrupts the connection between the communication unit **30** and the smartphone **50** and executes the disconnection process of the application. In the subsequent step S**4**, the communication control unit **12** sets a disconnection flag, which indicates whether the connection with the smartphone **50** is interrupted or not, to "1" (disconnected) and advances the process to step S**5**. In step S**20**, the communication control unit **12** sets the disconnection flag to "0" (not disconnected) and advances the process to step S**5**.

[0037] In step S**5**, the access point detection unit **11** searches for external APs **200** that can be connected to the communication unit **30** based on the radio wave reception condition of the communication unit **30** in the 2.4 GHz and 5 GHz bands. In the subsequent step S**6**, the access point detection unit **11** advances the process to step S**7** if the access point detection unit **11** detects external APs **200** that can be connected to the communication unit **30**, or to step S**30** if the access point detection unit **11** does not detect external APs **200** that can be connected to the communication unit **30**.

[0038] In step S**30**, the communication control unit **12** displays on the display device an announcement image **40** indicating that there are no connectable external APs **200**, and advances the process to step S**13** in FIG. **5**. In step S**7**, the communication control unit **12** determines whether the disconnection flag is "1" or not and advances the process to step S**8** if the disconnection flag is "1" or to step S**40** if the disconnection flag is "0."

[0039] In step S**8**, the communication control unit **12** displays on the display device **40** an access point information image **70** presenting a list of information on the external APs **200** detected by the access point detection unit **11** (AP-1, AP-2, and AP-3 are shown in the figure by way of example), as shown in FIG. **6**, and advances the process to step S**9** in FIG. **5**. The access point information image **70** presents information on AP-1 to AP-3 by means of selection buttons **71** to **73**, and the communication control unit **12** receives selection of one of the external APs **200** based on touch operation of one of the selection buttons by the occupant P.

[0040] In the access point information image **70**, the button **73** for AP-3, which supports communication connection in the 5.0 GHz band, is hatched to make the button **71** for AP-1 and the button **72** for AP-2, which support communication connection in the 2.4 GHz band, more viewable than the button **73** for AP-3. In this way, information on external APs **200** (AP-1 and AP-2 in this example) that support communication connection in the 2.4 GHz band, which do not overlap with the 5 GHz band used for the connection with the smartphone **50**, is presented preferentially over and differently from information on external APs **200** (AP-3 in this example) that support communication connection in the 5 GHz band, thereby encouraging the occupant P to select an external AP **200** that supports the 2.4 GHz band to prevent the connection with the smartphone **50** from being interrupted due to overlap between the frequency bands used.

[0041] In contrast, in step S**40**, the communication control unit **12** displays on the display device **40** an access point information image **80** that, equally and without distinction, presents buttons **81** and **82** indicating information on the external APs **200** that support the 2.4 GHz band and a button

83 indicating information on the external AP **200** that supports the 5 GHz band, as shown in FIG. **7**, and advances the process to step S**13** in FIG. **5**.

[0042] In step S**9** in FIG. **5**, the communication control unit **12** advances the process to step S**10** when it recognizes an operation to select one of the external APs **200** in the access point information image **70** (see FIG. **6**) or the access point information image **80** (see FIG. **7**) by the occupant P. In step S**10**, the communication control unit **12** determines whether the disconnection flag is "1" or not and advances the process to step S**11** if the disconnection flag is "1" or to step S**14** if the disconnection flag is "b **0**."

[0043] In step S**11**, the communication control unit **12** determines whether or not an external AP **200** that supports communication connection in the 5 GHz band is selected by the occupant P. The communication control unit **12** advances the process to step S**12** if an external AP **200** that supports communication connection in the 5 GHz band is selected, and to step S**50** if an external AP **200** that does not support communication connection in the 5 GHz band (but supports communication connection in the 2.4 GHz band) is selected.

[0044] In step S**12**, as shown in FIG. **6**, the communication control unit **12** displays on the display device **40** a communication unavailability announcement image **75** announcing that the connection with the smartphone **50** is interrupted and communication with the smartphone **50** becomes unavailable if AP-3, which is an external AP **200** that supports communication connection in the 5 GHz band, is selected. The announcement image **75** includes a continue button **76** and a cancel button **77**.

[0045] In step S**13**, the communication control unit **12** determines whether the continue button **76** is selected or not. The communication control unit **12** then advances the process to step S**14** if the continue button **76** is selected, or to step S**14** if the continue button **76** is not selected (if the cancel button **77** is selected).

[0046] In step S**50**, because the occupant P has selected an external AP **200** that supports communication connection in the 2.4 GHz band and thus the frequency band used with the smartphone **50** does not overlap, the communication control unit **12** reconnects the communication unit **30** and the smartphone **50** (reconnects in AP mode in the 5 GHZ band) and continues to execute the smartphone-linked application, and advances the process to step S**14**. In step S**14**, the communication control unit **12** connects the external AP **200** selected by occupant P and the communication unit **30** in STA mode.

4. Alternative Embodiments

[0047] In the above embodiment, the vehicle **100** is shown as an example of the mobile object in which the communication device **1** is used, but the mobile object in this disclosure may be any other type of mobile object in which occupants use wireless communication, such as an aircraft and a ship.

[0048] In the above embodiment, as an access point information image that presents information on external APs **200** that support communication connection in a second frequency band that does not overlap with a first frequency band used for communication between the communication unit **30** and the smartphone **50** differently from information on external APs **200** that support communication connection in the first frequency band, the display image **70** is shown, in which information on the external AP **200** that uses the first frequency band is hatched, as shown in FIG. **6**.

[0049] As alternative embodiments of such a display image, for example, an access point information image may be displayed in the display styles according to the following variants 1 through 5.

[0050] Variant 1—Only information on external APs **200** that support communication connection in the second frequency band is presented, and information on external APs **200** that support communication connection in the first frequency band is hidden.

[0051] Variant 2—As shown in FIG. **6**, the order of presenting information on external APs **200** is as shown in FIG. **6**, i.e., information on external APs **200** that support communication connection in the second frequency band is presented first from the top, followed by information on external APs **200** that support communication connection in the first frequency band.

[0052] Variation 3—The brightness or luminance of display of information on external APs **200** that support communication connection in the second frequency band is higher than that of information on external APs **200** that support communication connection in the first frequency band.

[0053] Variant 4—information on external APs **200** that support communication connection in the second frequency band and information on external APs **200** that support communication connection in the first frequency band are presented in different colors.

[0054] Variant 5—information on external APs **200** that support communication connection in the second frequency band is presented larger in size than information on external APs **200** that support communication connection in the first frequency band.

[0055] The display style of the access point information image may be changed to one of the display styles shown in FIG. **6** and the above variants 1 to 5 according to an operation to set the display style of the access point information image by the occupant P.

[0056] In the above embodiment, in step S**12** in FIG. **5**, the communication control unit **12** displays on the display device **40** a communication unavailability announcement image **75** announcing that communication with the smartphone **50** becomes unavailable if an external AP **200** that supports the 5 GHz band is selected by the occupant P. As an alternative embodiment, instead of or together with the communication unavailability announcement image **75**, a recommended frequency band announcement image may be displayed on the display device **40**, which recommends connection with an external AP **200** that supports communication connection in the 2.4 GHz band, which does not overlap with the frequency band used for the smartphone **50**. The communication unavailability announcement and the recommended frequency announcement may be performed by audio output from the on-board speaker **42**.

[0057] In the above embodiment, the communication device **1** installed in the vehicle **100** is shown, but the communication device in the present disclosure may also be a portable communication device that is carried and used in a mobile object. In this case, the communication device may be a display-integrated communication device including a display device.

[0058] FIG. **2** is a schematic diagram illustrating the configuration of the communication device **1** divided according to the main processing features to facilitate understanding of the present invention, but the configuration of the communication device **1** may be divided in different ways. The processing of each component may be performed by a single hardware unit or by multiple hardware units. The process executed by each component shown in FIGS. **4** and **5** may be performed by a single program or by multiple programs.

5. Configurations Supported by Above Embodiment

[0059] The above embodiment is a specific example of the following configurations.

[0060] (Configuration 1) A communication device used in a mobile object, including: a communication unit configured to perform wireless communication with communication targets in a first frequency band or a second frequency band; an access point detection unit configured to detect access points capable of communication connection with the communication unit; and a communication control unit configured to control communication between the communication unit and a communication terminal and communication between the communication unit and the access points, and to display an access point information image on a display device used in the mobile object when the communication between the communication unit and the communication terminal is performed in the first frequency band, the access point information image presenting information on the access points detected by the access point detection unit, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in the second frequency band.

[0061] With the communication device in configuration **1**, an access point information image is displayed on the display device, which presents information on access points capable of communication with the communication unit, with a first access point capable of communication in the first frequency band being presented differently from a second access point capable of communication in the second frequency band. This can make an occupant of the mobile object recognize that there is a second access point capable of communication connection in the second frequency band that does not overlap with the first frequency band in which communication with the communication terminal is being performed, thereby encouraging the occupant to perform communication with the second access point in the second frequency band to prevent the communication with the communication terminal from being interrupted by connection with an access point during the communication with the communication terminal.

[0062] (Configuration 2) The communication device according to configuration 1, wherein the communication control unit is further configured to display the access point information image on the display device when the communication between the communication unit and the communication terminal is performed in the first frequency band and the communication control unit recognizes an operation to connect the communication unit and the access points by an occupant of the mobile object.

[0063] With the communication device in configuration **2**, by displaying the access point information image on the display device at the timing of the occupant of the mobile object performing an operation to connect to an access point, information on the second access point capable of connection in the second frequency band that does not overlap with the first frequency band used for communication with the communication terminal can be effectively provided to the occupant of the mobile object.

[0064] (Configuration 3) The communication device according to configuration 1 or configuration 2, wherein the communication control unit is further configured to interrupt the communication between the communication unit and the communication terminal in the first frequency band when the communication between the communication unit and the communication terminal is performed in the first frequency band and the communication control unit recognizes an operation to connect the communication unit and the access points by an occupant of the mobile object, and the access point detection unit is further configured to detect the access points capable of communication connection with the communication unit by performing communication in the first frequency band and the second frequency band while the communication between the communication unit and the communication terminal is interrupted.

[0065] With the communication device in configuration 3, the communication between the communication unit and the communication terminal can be interrupted, and the access point detection unit can perform a search in the first frequency band and the second frequency band to recognize access points capable of connection with the communication unit.

[0066] (Configuration 4) The communication device according to configuration 3, wherein the communication control unit is further configured to start communication between the communication unit and the second access point in the second frequency band and resume the communication between the communication unit and the communication terminal in the first frequency band when the communication control unit recognizes an operation to connect to the second access point by the occupant in response to display of the access point information image on the display device.

[0067] With the communication device in configuration 4, both communication between the communication unit and the communication terminal in the first frequency band and communication between the communication unit and the second access point in the second frequency band can be performed in response to an operation to connect to the second access point by the occupant.

[0068] (Configuration 5) The communication device according to any one of configurations 1 to 4, wherein the communication control unit is further configured to output a communication unavailability announcement that communication with the terminal device becomes unavailable from an announcement device used in the mobile object when the communication control unit recognizes an operation to connect to the first access point by an occupant of the mobile object in response to display of the access point information image on the display device.

[0069] With the communication device in configuration 5, by announcing to the occupant who performs an operation to connect to the first access point that the communication with the communication terminal becomes unavailable, the occupant can be encouraged to connect to the second access point to prevent the communication between the communication unit and the communication terminal from being interrupted.

[0070] (Configuration 6) The communication device according to any one of configurations 1 to 5, wherein the communication control unit is further configured to output a recommended frequency band announcement recommending connection with the second access point from an announcement device used in the mobile object when the communication control unit recognizes an operation to connect to the first access point by an occupant of the mobile object in response to display of the access point information image on the display device.

[0071] With the communication device in configuration 6, by announcement recommending the occupant who performs an operation to connect to the first access point to perform communication with communication terminal in the second frequency band, the occupant can be encouraged to continue the communication between the communication unit and the communication terminal by changing selection of the frequency band used for the communication between the communication unit and the communication terminal from the first frequency band to the second frequency band.

[0072] (Configuration 7) The communication device according to any one of configurations 1 to 6, wherein the communication control unit is further configured to display on the display device as the access point information image an image in which only information on the second access point is presented and information on the first access point is hidden.

[0073] With the communication device in configuration 7, by presenting only information on the second access point and hiding information on the first access point in the access point information image, the occupant can be encouraged to connect to the second access point to prevent the communication between the communication unit and the communication terminal from being interrupted due to overlap between the frequency bands used.

[0074] (Configuration 8) The communication device according to any one of configurations 1 to 6, wherein the communication control unit is further configured to display on the display device as the access point information image an image in which information on the second access point is presented preferentially over information on the first access point.

[0075] With the communication device in configuration 8, by presenting information on the second access point preferentially over information on the first access point in the access point information image, the occupant can be encouraged to connect to the second access point to prevent the communication between the communication unit and the communication terminal from being interrupted due to overlap between the frequency bands used.

[0076] (Configuration 9) The communication device according to any one of configurations 1 to 8, wherein the communication control unit is further configured to change a display style of the access point information image according to an operation to set the display style of the access point information image by an occupant of the mobile object.

[0077] With the communication device in configuration 9, the access point information image can be changed to a display style that suits the occupant's preference according to a setting operation by the occupant, thereby facilitating selection of an access point by the occupant.

[0078] (Configuration 10) A communication control method performed by a communication device used in a mobile object and including a communication unit configured to perform wireless communication with communication targets in a first frequency band or a second frequency band, the method including: an access point detection step of detecting access points capable of communication connection with the communication unit; and a communication control step of controlling communication between the communication unit and a communication terminal and communication between the communication unit and the access points, and displaying an access point information image on a display device used in the mobile object when the communication between the communication unit and the communication terminal is performed in the first frequency band, the access point information image presenting information on the access points detected in the access point detection step, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in the second frequency band.

[0079] By performing the communication control method in configuration **10** by a communication device, the same effect as the communication device in configuration **1** can be achieved.

[0080] **1** . . . communication device, **10** . . . processor, **11** . . . access point detection unit, **12** . . . communication control unit, **20** . . . memory, **21** . . . control program for communication device, **22** . . . smartphone-linked application program, **23** . . . assistant application program, **30** . . . communication unit, **40** . . . display device, **41** . . . on-board microphone, **42** . . . on-board speaker, **50** . . . smartphone (communication terminal), **70**, **80**. . . . AP list image, **71-73**, **81-83**. . . . AP information button, **75** . . . connection interruption announcement image, **100** . . . vehicle (mobile object), **200** . . . external AP, **210** . . . communication network, P . . . occupant.

1. A communication device used in a mobile object, comprising:

a communication unit configured to perform wireless communication with communication targets in a first frequency band or a second frequency band;

an access point detection unit configured to detect access points capable of communication connection with the communication unit; and

a communication control unit configured to control communication between the communication unit and a communication terminal and communication between the communication unit and the access points, and to display an access point information image on a display device used in the mobile object when the communication between the communication unit and the communication terminal is performed in the first frequency band, the access point information image presenting information on the access points detected by the access point detection unit, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in the second frequency band.

2. The communication device according to claim **1**, wherein the communication control unit is further configured to display the access point information image on the display device when the communication between the communication unit and the communication terminal is performed in the first frequency band and the communication control unit recognizes an operation to connect the communication unit and the access points by an occupant of the mobile object.

3. The communication device according to claim **1**, wherein

the communication control unit is further configured to interrupt the communication between the communication unit and the communication terminal in the first frequency band when the communication between the communication unit and the communication terminal is performed in the first frequency band and the communication control unit recognizes an operation to connect the communication unit and the access points by an occupant of the mobile object, and

the access point detection unit is further configured to detect the access points capable of communication connection with the communication unit by performing communication in the first frequency band and the second frequency band while the communication between the communication unit and the communication terminal is interrupted.

4. The communication device according to claim **3**, wherein the communication control unit is further configured to start communication between the communication unit and the second access point in the second frequency band and resume the communication between the communication unit and the communication terminal in the first frequency band when the communication control unit recognizes an operation to connect to the second access point by the occupant in response to display of the access point information image on the display device.

5. The communication device according to claim **1**, wherein the communication control unit is further configured to output a communication unavailability announcement that communication with the terminal device becomes unavailable from an announcement device used in the mobile object when the communication control unit recognizes an operation to connect to the first access point by an occupant of the mobile object in response to display of the access point information image on the display device.

6. The communication device according to claim **1**, wherein the communication control unit is further configured to output a recommended frequency band announcement recommending connection with the second access point from an announcement device used in the mobile object when the communication control unit recognizes an operation to connect to the first access point by an occupant of the mobile object in response to display of the access point information image on the display device.

7. The communication device according to claim **1**, wherein the communication control unit is further configured to display on the display device as the access point information image an image in which only information on the second access point is presented and information on the first access point is hidden.

8. The communication device according to claim **1**, wherein the communication control unit is further configured to display on the display device as the access point information image an image in which information on the second access point is presented preferentially over information on the first access point.

9. The communication device according to claim **1**, wherein the communication control unit is further configured to change a display style of the access point information image according to an operation to set the display style of the access point information image by an occupant of the mobile object.

10. A communication control method performed by a communication device used in a mobile object and including a communication unit configured to perform wireless communication with communication targets in a first frequency band or a second frequency band, the method comprising:

an access point detection step of detecting access points capable of communication connection with the communication unit; and

a communication control step of controlling communication between the communication unit and a communication terminal and communication between the communication unit and the access points, and displaying an access point information image on a display device used in the mobile object when the communication between the communication unit and the communication terminal is performed in the first frequency band, the access point information image presenting information on the access points detected in the access point detection step, with a first access point capable of communication connection in the first frequency band being presented differently from a second access point capable of communication connection in the second frequency band.

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